

Zero weak gradient gives zero derivatives of local mollifications

MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero

theorem

Substitutes a translated smooth kernel into the weak equation and tracks the reflection sign.

Statement

Let E be a finite-dimensional real normed vector space with its Borel structure and an additive Haar measure μ . Let $U \subseteq E$ be open and $u : E \rightarrow \mathbb{R}$ locally integrable on U . Assume

$$\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$$

for every C^1 vector field $W : E \rightarrow E$ for which a compact set K_W satisfies

$$\overline{\{z \in E : W(z) \neq 0\}} \subseteq K_W \subseteq U.$$

Here $\operatorname{div} W(z) := \operatorname{tr}(DW(z))$. Fix $x \in E$ and $R > 0$ with $K := \overline{B}(x, 4R) \subseteq U$, and define

$$v(z) := \begin{cases} u(z), & z \in K, \\ 0, & z \notin K. \end{cases}$$

For $k \in \mathbb{N}$, let φ_k be the normalized smooth bump centered at zero with inner radius $R/(2(k+2))$ and outer radius $r_k = R/(k+2)$; thus $\varphi_k \geq 0$, $\int_E \varphi_k d\mu = 1$, and its closed support lies in $\overline{B}(0, r_k)$. Put

$$v_k(y) := \int_E \varphi_k(t) v(y - t) d\mu(t).$$

Then for every $y \in B(x, R)$, every $k \in \mathbb{N}$, and every $e \in E$,

$$Dv_k(y)e = 0.$$

Assumptions

The ball data are supplied here; the theorem does not choose them. Only local integrability of u on U is required, not global integrability or Lipschitz continuity. The weak predicate alone would not imply integrability, so the separate local-integrability hypothesis is retained. Dimension zero and any direction, including $e = 0$, are allowed.

Conclusion

Every directional derivative of this localized mollification vanishes at every point of the inner ball. The conclusion is pointwise for the smooth function v_k ; it does not claim that the rough function u is pointwise constant.

The local ball data also name the middle ball $\overline{B}(x, 2R)$. The translated test field used in the proof has a chosen compact carrier inside this middle ball and hence inside $K \subseteq U$. A carrier is a compact set containing the nonzero set and its closure; it need not equal either. For a set S , $\mathbf{1}_S$ denotes its indicator, equal to one on S and zero off S . Thus the localized function is also written $v = \mathbf{1}_K u$.

Proof route

Use $W(z) = \varphi_k(y - z)e$ as the actual vector test field. Its divergence is the negative kernel derivative. Compact localization leaves its weak integral unchanged, and the derivative formula for convolution identifies the resulting zero integral with $Dv_k(y)e$.

▼ Proof steps

1. Construct an admissible field. Define

$$W(z) := \varphi_k(y - z)e, \quad K_W := \overline{B}(y, r_k).$$

It is C^1 (in fact smooth), and its nonzero set is contained in K_W . Finite dimensionality makes this closed ball compact. For $z \in K_W$, since $\|y - x\| < R$ and $r_k \leq R/2$,

$$\|z - x\| \leq \|z - y\| + \|y - x\| < r_k + R \leq \frac{3}{2}R < 2R.$$

Thus $K_W \subseteq \overline{B}(x, 2R) \subseteq K \subseteq U$, so the weak equation applies to this very field and carrier.

2. Compute the divergence with its sign. For an arbitrary direction $h \in E$, differentiating the reflected smooth kernel gives

$$DW(z)h = (D\varphi_k(y - z)(-h))e = -(D\varphi_k(y - z)h)e.$$

The rank-one trace identity says that the trace of the map $h \mapsto L(h)e$ is $L(e)$. Substituting $L = D\varphi_k(y - z)$ gives

$$\operatorname{div} W(z) = -D\varphi_k(y - z)e.$$

The minus sign comes from differentiating $z \mapsto y - z$.

3. Identify the localized integral. On K_W , both $\mathbf{1}_K u = u$ and $\mathbf{1}_U = 1$. Off K_W , the field is zero on an open neighborhood, because K_W is closed and contains its nonzero set. Consequently $DW = 0$ and $\operatorname{div} W = 0$ there. These two cases give the pointwise identity

$$v(z) \operatorname{div} W(z) = \mathbf{1}_U(z) u(z) \operatorname{div} W(z).$$

As U is open and hence measurable, integration and the weak equation yield

$$\int_E v(z) \operatorname{div} W(z) d\mu(z) = \int_U u(z) \operatorname{div} W(z) d\mu(z) = 0.$$

Here the compact-localization integrability lemma applies to the given local integrability on U and the compact set $K \subseteq U$, and makes v globally integrable. The kernel derivative is continuous, bounded, and compactly supported after translation; the displayed products are integrable as well.

4. **Substitute the derivative formula.** The localized function v is locally integrable on E . Apply the derivative formula for local mollifications with this same v , the positive radius R , the index k , the point y , and the direction e . It identifies the last integral below with $Dv_k(y)e$. Using Step 2 in Step 3 gives

$$\begin{aligned} 0 &= \int_E v(z) [-D\varphi_k(y-z)e] d\mu(z) \\ &= - \int_E v(z) D\varphi_k(y-z)e d\mu(z) = -Dv_k(y)e. \end{aligned}$$

Therefore $Dv_k(y)e = 0$. The calculation differentiates the smooth kernel and its convolution, never u or the discontinuous cutoff $\mathbf{1}_K u$.

Main citations

- Exact declaration and its source — `MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`
- Weak test equation with chosen compact carriers — `MathlibAnnex.WeakDivergenceZero`
- Derivative of convolution through the compact kernel — `MathlibAnnex.WeakGradient.localMollification_fderiv_apply`
- Rank-one trace identity — `MathlibAnnex.trace_smulRight`
- Local ball data and the three nested balls — `MathlibAnnex.WeakGradient.LocalBallData`
- Compact localization of a locally integrable function — `MathlibAnnex.WeakGradient.integrable_compactLocalization`
- Divergence of a reflected scalar kernel times a direction — `MathlibAnnex.divergence_sub_smul`
- Derivative vanishes on a zero neighborhood off the carrier — `MathlibAnnex.CompactC1VectorField.fderiv_eq_zero_of_not_mem_carrier`
- Translated small closed balls lie in the middle ball — `MathlibAnnex.WeakGradient.LocalBallData.translated_closedBall_subset_middle`
- The outer radius is at most half the local radius — `MathlibAnnex.WeakGradient.shrinkingBump_rOut_le_half`
- The translated bump vector test field — `MathlibAnnex.WeakGradient.translatedBumpField`
- Its compact carrier lies in the open domain — `MathlibAnnex.WeakGradient.translatedBumpField_carrier_subset`
- Divergence of the translated bump field — `MathlibAnnex.WeakGradient.divergence_translatedBumpField`
- The weak integral equals the localized whole-space integral — `MathlibAnnex.WeakGradient.weakIntegral_eq_localized_bumpIntegral`

Lean source signature (exact)

```
theorem localMollification_fderiv_apply_eq_zero
  {U : Set E} {u : E → ℝ}
  (hU : IsOpen U)
  (hu : LocallyIntegrableOn u U μ)
  (hweak : WeakDivergenceZero μ U u)
  {x : E} (D : LocalBallData U x)
```

```

{y : E} (hy : y ∈ D.inner) (k : ℕ) (e : E) :
fderiv ℝ
  (localMollification (μ := μ) D.radius D.radius_pos
    (compactLocalization D.carrier u) k) y (e) = 0

```

In the source	Mathematical meaning
<pre> [NormedAddCommGroup E] [NormedSpace ℝ E] [FiniteDimensional ℝ E] [MeasurableSpace E] [BorelSpace E] {μ : Measure E} [μ.IsAddHaarMeasure] </pre>	The finite-dimensional real normed space E, with Borel measurable structure and additive Haar measure μ. Dimension zero is allowed.
<pre> (hweak : WeakDivergenceZero μ U u) </pre>	The scalar weak equation $\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$ for every C^1 vector field with a compact carrier contained in U; $\operatorname{div} W = \operatorname{tr}(DW)$.
<pre> (hU : IsOpen U) (hu : LocallyIntegrableOn u U μ) </pre>	The set U is open and u is locally integrable on U. Whole-space integrability or Lipschitz continuity of u is not assumed.
<pre> {x : E} (D : LocalBallData U x) </pre>	The point x and the supplied ball record. Its three fields are read in the next three rows.
<pre> D.radius </pre>	The radius $R \in \mathbb{R}$.
<pre> D.radius_pos </pre>	The hypothesis $R > 0$.
<pre> D.closed_four_subset </pre>	The closed ball $\overline{B}(x, 4R)$ is contained in U, with this same x, R.
<pre> D.inner; D.middle; D.carrier </pre>	The cited definitions give $B(x, R)$, $\overline{B}(x, 2R)$ and $K = \overline{B}(x, 4R)$, respectively.
<pre> {y : E} (hy : y ∈ D.inner) (k : ℕ) (e : E) </pre>	The point $y \in B(x, R)$, kernel index k and arbitrary direction e.
<pre> compactLocalization D.carrier u </pre>	The scalar function $v = \mathbf{1}_K u$, equal to u on K and zero elsewhere.
In the source	Mathematical meaning
<pre> localMollification (μ := μ) D.radius D.radius_pos (compactLocalization D.carrier u) k </pre>	The same localized mollification v_k, with the normalized kernel of radii $R/(2(k+2))$ and $R/(k+2)$.
<pre> fderiv ℝ (localMollification (μ := μ) D.radius D.radius_pos (compactLocalization D.carrier u) k) y (e) = 0 </pre>	The conclusion $Dv_k(y)e = 0$, pointwise for this smooth function. It does not assert a derivative or pointwise constancy of u.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 11–17:

```
noncomputable section
open Set Metric MeasureTheory Filter TopologicalSpace ContinuousLinearMap
open scoped ENNReal Topology Convolution BigOperators
namespace MathlibAnnex.WeakGradient
variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [FiniteDimensional ℝ E] [MeasurableSpace E] [BorelSpace E]
variable {μ : Measure E} [μ.IsAddHaarMeasure]
```

▼ Exact content identity

Declaration: `MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`

Accepted content SHA-256: `5c79f9b268f22ebca5088199373a63f4e1f0103fe21e49a328326e591727acbd`

Accepted source guide SHA-256: `2ec6c53eb4326bb445202980fed27249a05c34c9e9480a01cbeac8d3e444fe3a`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`